GHG Emissions Data Dictionary & Documentation

Data Dictionary — Nova Scotia GHG Emissions Dataset

This dataset contains cleaned greenhouse gas (GHG) emissions data specifically for Nova Scotia. It is structured for time-series analysis, sector-level insights, and predictive modelling.

Column	Type	Description
Year	int64	Reporting year
Emissions	float64	GHG emissions (kt CO_2e)
Sectors	object	Emission sector

Data Dictionary — Canada Province-Level GHG Dataset

This dataset includes greenhouse gas emissions across all Canadian provinces and territories for comparative and benchmarking analysis.

Column	Type	Description
Year	int64	Reporting year
Emissions	float64	GHG emissions (kt CO_2e)
Provinces	object	Province name
Sectors	object	Emission sector

Project Methodology & Documentation

Data Source

Government of Canada Open Data and Statistics Canada.

Data Preparation

The raw dataset was processed using Python (Pandas):

- Removed unnecessary metadata columns

- Filtered Nova Scotia data into a separate dataset
- Standardized column names
- Converted data types (Year → int, Emissions → float)
- Removed duplicates and null values
- Cleaned sector labels by removing embedded classification codes

Dataset Structuring

Two datasets were created:

- **Nova Scotia dataset (df_ns)**
  Used for detailed analysis and forecasting
- **Province dataset (df_province)**
  Used for comparative and benchmarking analysis

Exploratory Data Analysis (EDA)

Trend analysis, KPI calculation, sector analysis, and province comparisons.

EDA was conducted to answer key analytical questions:

- How have GHG emissions changed over time in Nova Scotia?
- Which sectors contribute most to emissions?
- How does Nova Scotia compare to national trends?
- Which provinces show the greatest change?

Analysis included:

- Time-series trend visualization
- KPI calculation (percentage change)
- Sector-level aggregation
- Province-level comparison

Predictive Modelling

A **linear regression model** was applied to forecast future emissions trends.

- Independent variable: Year
- Dependent variable: Emissions

The model was used to predict emissions for the next 5 years.

Assumptions

Linear trend assumed; external factors not included.

- Emissions follow a linear trend over time
- External factors (policy, economic changes) are not included
- Forecast results are indicative, not definitive

Outputs

Results were exported for Power BI and include:

- Time-series dataset
- Sector-level dataset
- Forecast dataset
- KPI metrics

Limitations

Aggregated data and simple model limit accuracy.

- Aggregated sector-level data may hide finer details
- Forecast model is simplified
- External variables not considered